Library User Manual

AXL library family
AXL

AJINEXTEK AXT

Product Information

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AXL Overview

Control board of Ajinextek can have various formats depending on how to combine motion module, digital I/O module, analog I/O module, and communication module into base board and having versatile functions.

As it is a combination of single boards of specific functions, not a module format, it can be made into a format showing excellent functions and cost effective. Single board can be understood as Module mounted on board to achieve a unified method for using library. In other worlds, single board is similar to module format that has module mounted on base board.

The program library API (Application Programming Interface) that controls Ajinextek's boards and modules is called AXL (Ajinextek Library).

AXL products are available as follows.

Product	Feature	Note
BPFR(Full size)	Module mounted on base board (PCI bus)	Up to 4 modules
BPHR(Half size)	Module mounted on base board (PCI bus)	Up to 2 modules
BIFR(Full size)	Module mounted on base board (ISA bus)	Up to 4 modules
BIHR(Half size)	Module mounted on base board (ISA bus)	Up to 2 modules
BPHD(Half size)	Module mounted on base board, DB32P default (PCI bus)	Up to 2 modules, Digital Input (16)/ Output (16) default
PCI-DI64x	Integrated digital input board (PCI bus)	Digital Input 64
PCI-DB64x	Integrated digital input board (PCI bus)	Digital Input 32/ Digital Output 32
PCI-DO64x	Integrated digital input board (PCI bus)	Digital Output 64
PCI-AIO1602HR	Integrated analog input board (PCI bus)	Analog Output (2CH), Input (16CH)
PCI-N804(Half size)	Integrated motion control board (PCI bus)	8 controlled axis, Open loop
PCI-N404(Half size)	Integrated motion control board (PCI bus)	4 controlled axis, Open loop
PCI-R1604	Network master board (PCI bus)	RTEX(Real-Time Express) network master board up to 32 slave nodes, basicly 16 axes, 32 axes controlled expansion.
PCI-R1604-MLII	Network master board (PCI bus)	Mechatrolink-II network master board up to 30 slave nodes, basicly 16 axes, extended 30 axis

AXL base board-mounted modules are available as follows.

Product	Feature	Note
SMC-2V03	Motion module	2 controlled axis,Pulse Output, (Obsolete)
SMC-2V04	Motion module	2 controlled axis,Pulse Output
SIO-DI32	Digital Input module	Digital Input 32 channel
SIO-DB32	Digital Input/Output module	Digital Input(16)/Output(16) channel
SIO-DO32	Digital Output module	Digital Output 32 channel
SIO-AI4RB	Analog Input module	Analog Input 4CH
SIO-AO4RB	Analog Output module	Analog Output 4CH

AXL slave nodes are on the network that can be used as follows.

Product	Feature	Note
RTEX-PM	RTEX Motion Slave Node	1 axis, pulse output (Ajinextek)
RTEX-DI32	RTEX Digital Input Slave Node	Digital Input 32 channel (Ajinextek)
RTEX-DO32	RTEX Digital Output Slave Node	Digital Output 32 channel (Ajinextek)
RTEX-AI8	RTEX Analog Input Slave Node	Analog Input, 8CH (Ajinextek)
RTEX-AO4	RTEX Analog Output Slave Node	Analog Output, 4CH (Ajinextek)
RTEX-A4N	RTEX Motion Slave Node	1 axis, servo driver (Panasonic)
JEPMC-PL2910	Mechatrolink-II Motion Slave Node	2 axis, Pulse Output (Yaskawa)
JEPMC-AN2900	Mechatrolink-II Analog Input Slave Node	Analog Input, 4CH (Yaskawa)
JEPMC-AN2910	Mechatrolink-II Analog Output Slave Node	Analog Output, 2CH (Yaskawa)
JEPMC-IO2330	Mechatrolink-II Digital Input/Output Slave Node	Digital Input/Output 64/64 channel (Yaskawa)
SGDV-xxxx11A	Mechatrolink-II Motion Slave Node	1 axis, servo driver (Panasonic)

AXL library can be divided into the following 4 parts: AXL which is related to basic APIs, AMX which is related to Motion, AXD which is related to AXD, and AXA which is related to Analog VO. Library can be categorized by header file as follows.

AXL (Base Command)

AXL is a basic library that automatically recognizes, initializes, creates interface with all mounted boards and modules.

AXM (Motion Command)

AXM is a library related to motion control. It provides higher advanced motion APIs that include linear or circular interpolation function which is provided by chip. It also gives interface that makes programming easier. It only supports hardware higher than SMC-2V03 version that has motion control chip specified for double axes. For the case that lower hardware such as SMC-xV01 and SMC-xV02 can be used, a lower version library, AXT Library, can also be used.

AXD (DIO Command)

AXD is a library that controls input/output of digital signal through DIO module and board.

AXA (AIO Command)

AXA is a library that controls the conversion of analog signal to digital signal or the conversion of digital signal to analog signal through AIO module and board.

OS development environments that AXL library supports are as follows.

OS	Detail
Windows 98	Depending on user's environment that can support
Windows 2000	SP4 or later
Windows XP(32-bit)	SP3 or later
Windows XP(32-bit) + RTX kernnel	Windows XP(SP3 or later), RTX(7.0 or later)
Windows XP(64-bit)	Support

Windows 7(32-bit)	Professional or later
Windows 7(64-bit)	Professional or later

*. Windows 98/2000/XP/7(Microsoft of OS), RTX(IntervalZero of Real-Time extension kernel)

The development languages that AXL library supports are as follows.

Development Language	Detail
Visual C++	Visual studio 6.0, 2003, 2005, 2008, 2010
Visual Basic	Visual studio 6.0, 2003, 2005, 2008, 2010
C#	Visual studio 2005, 2008, 2010
C++Builder	6.0 or later
Delphi	7.0 or later

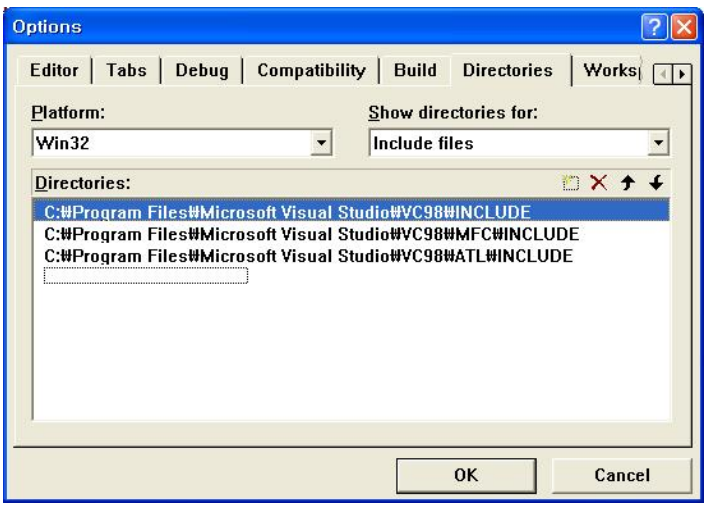
Library Initialization

Setting Library Path

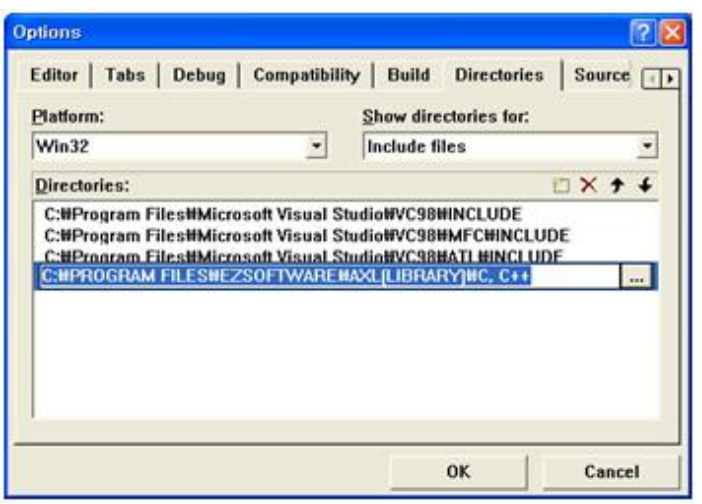
If library path is set in each language, it is unnecessary to set it again every time executing new project.

● *Linking Visual C++ library*

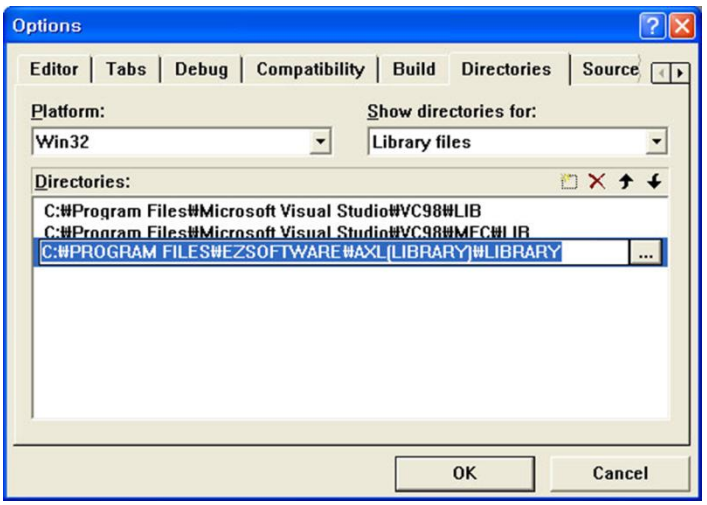
Run Visual C++. Click Tools -> Options in system menu to move to Directories tab. Designate Header file and Library file.



To designate Header file, select Include files in Show directories for: and click the last line in Directories. ... button will appear on the right. Click the button to pop up Choose Directory dialogue box. Designate ₩EzSoftware AXL₩AXL(Library)₩C folder.



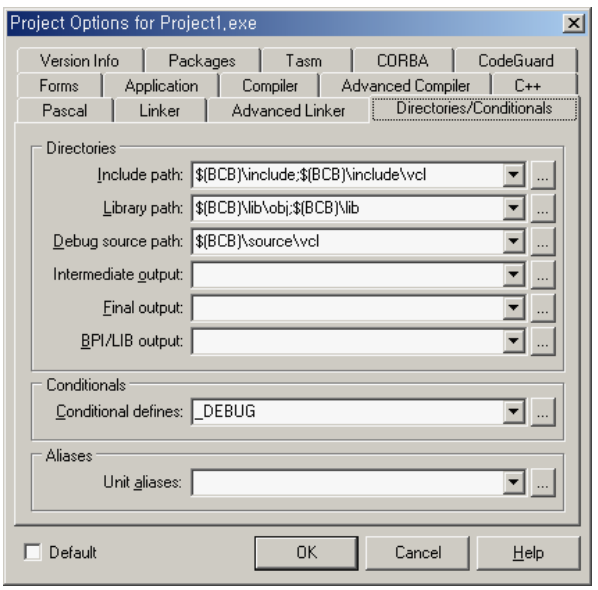
To designate Library file, select Library files in Show directories for: and click the last line in Directories. ... button will appear on the right. Click the button to pop up Choose Directory dialogue box. Designate ₩EzSoftware AXT₩AXL(Library)₩Library folder.



Click OK button if designating Header file and Library file is finished.

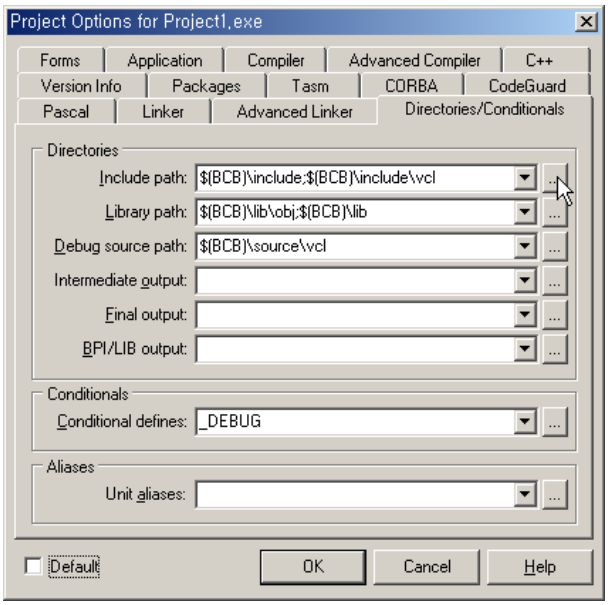
● *Linking C++ Builder library*

Run C++ Builder. Click Project -> Options in system menu to move to Directories/ Conditionals tab. Designate Header file and Library file.



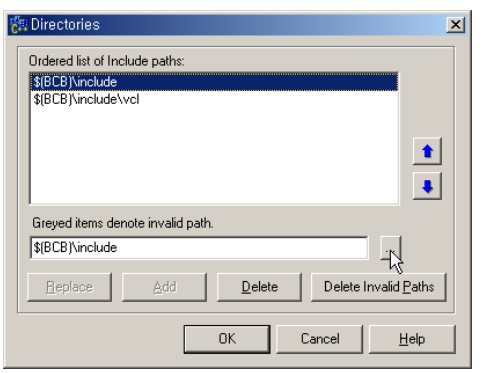
To designate Header file, click ... button on the right in Include path: item of Directories to pop up

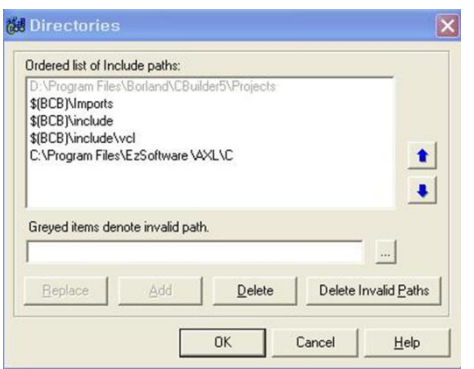
Directories dialogue box.



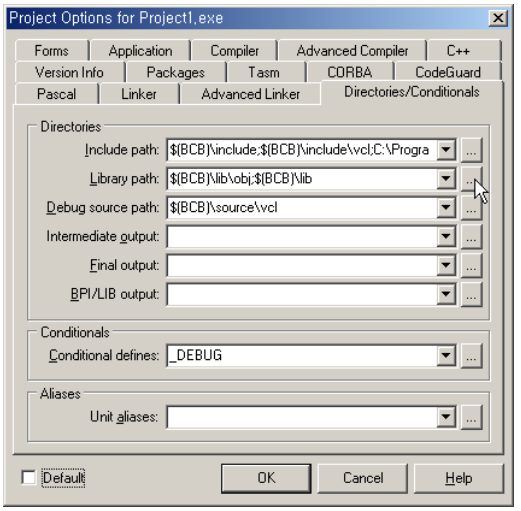
Click … button on the right in 'Greyed items denote invalid path' item to pop up dialogue box to find search.

Designate ₩EzSoftware AXL₩AXL(Library)₩C folder and click OK button.



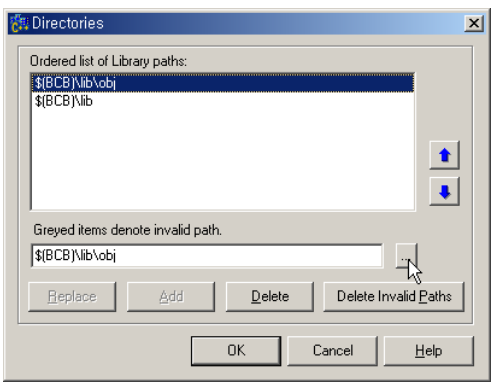


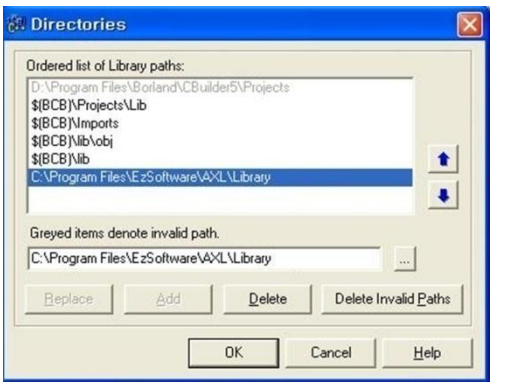
To designate Library file, click … button on the right in 'Library path:' item in Directories to pop up Directories dialogue box.



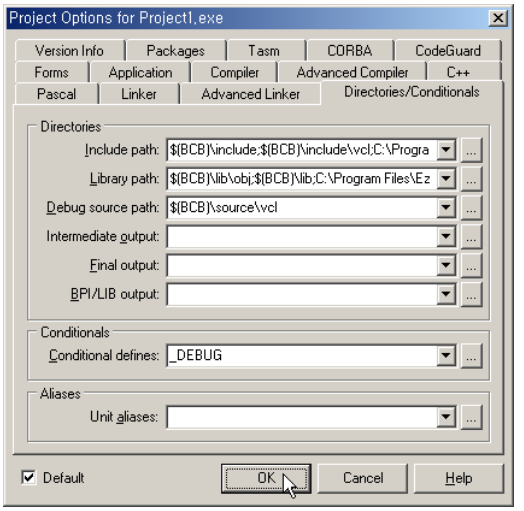
Click ... button on the right in 'Greyed items denote invalid path.' item to pop up dialogue box to search for folder.

Designate ₩EzSoftware AXL₩AXL(Library)₩Library folder and click OK button.



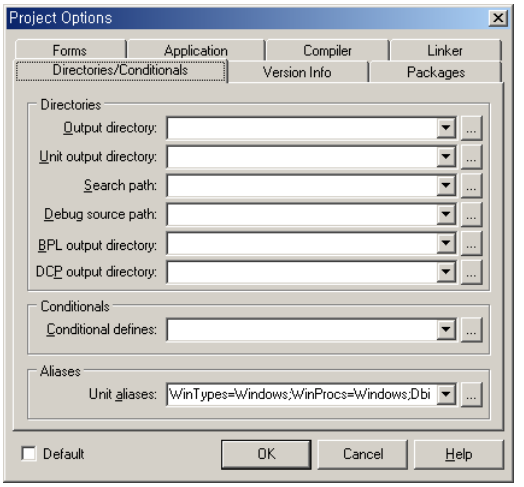


Check Default at the lower left corner and click OK button if designating Header file and Library file is finished.

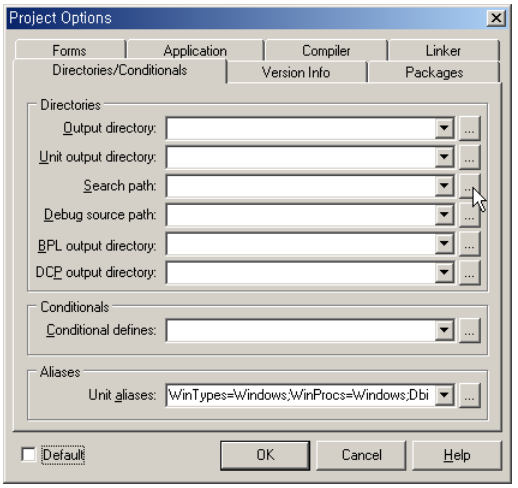


● *Linking Delphi library*

Run Delphi. Click Project -> Options in system menu to move to Directories/ Conditionals tab. Designate Header file and Library file.

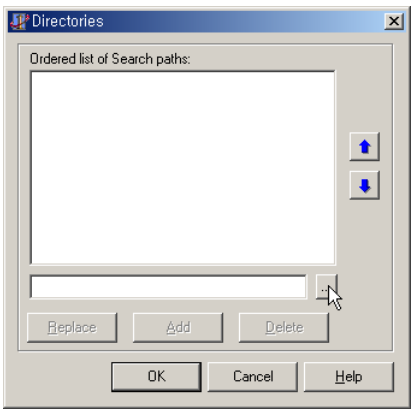


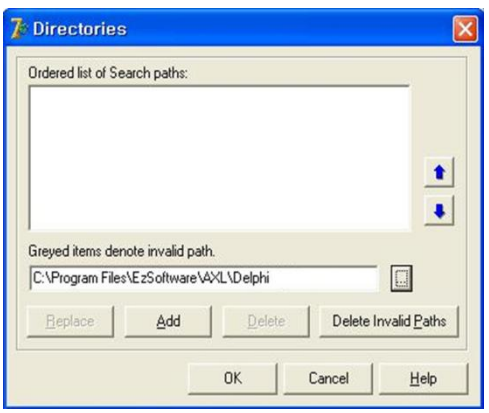
To designate Header file, click ... button on the right of 'Search path:' item in Directories to pop up Directories dialogue box.



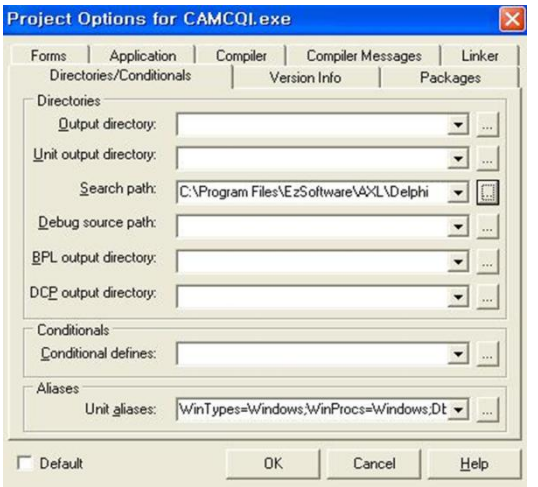
Click ... button on the right of 'Greyed items denote invalid path' item to make dialogue box to search for folder appear.

Designate ₩EzSoftware AXL₩AXL(Library)₩Delphi folder and click OK button.



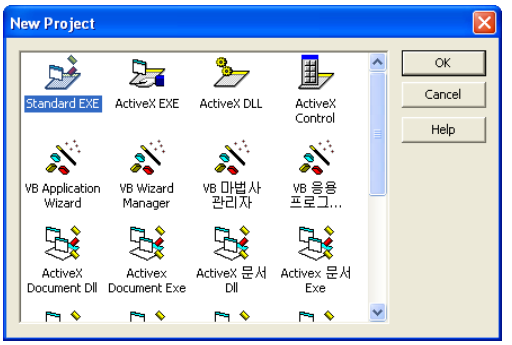


Check Default at the lower left corner and click OK button if designating Header file is finished.

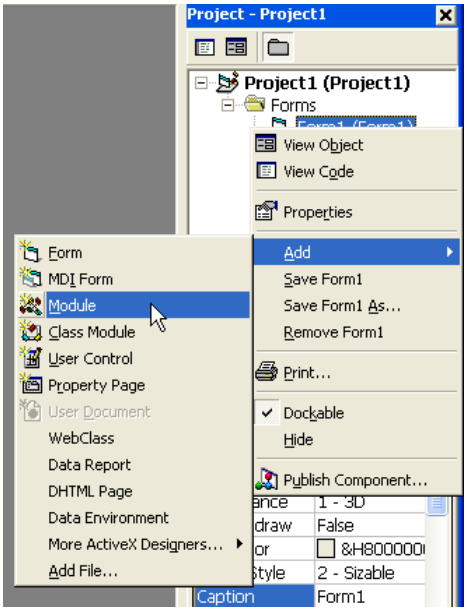


● *Linking Visual Basic library*

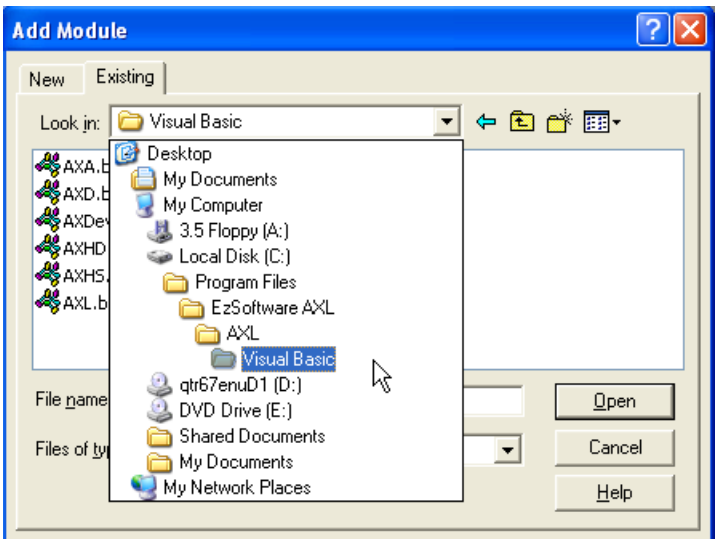
Run Visual Basic. Click File(F)-> New Project(N) in system menu, select Standard EXE, and click Confirm button.

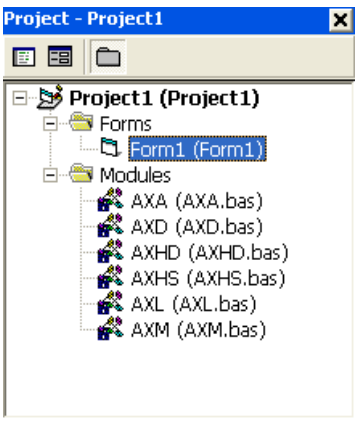


In project window on the right, click the right button of the mouse and select Add(A)->Module(M).



Move the old file tab. Designate ₩EzSoftware AXL₩AXL ₩Visual Basic folder by clicking Location(I).





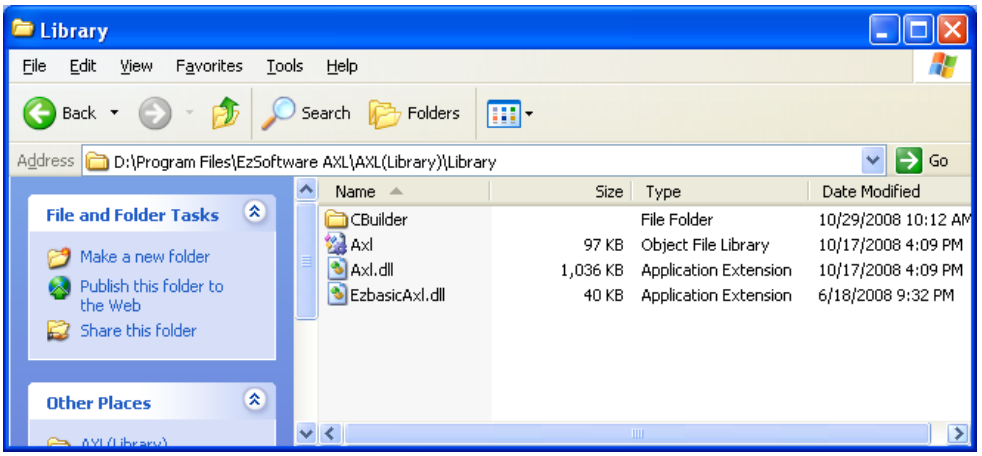
Open all modules. Repeat opening module since one module opens at a time.

The corresponding API can be used only after linking module.

DLL File Copy

If linking operation for a language is finished, AXL.dll, EzBasicAXL.dll must be in a folder that has execution file created by project compile.

User can copy file which is in ₩EzSoftware AXL₩AXL(Library) ₩Library. Yet, for Visual Basic, copy it to system folder of the OS. For example, if OS is Windows 2000, the folder is ₩WINNT₩system32.



Methods to Allot Axis Number and Channel Number

Basic Principle

1. *The same module is numbered at the same time.*

   As AXL only supports versions higher than 2V03 module, axis containing 2V02 is not recognized. For versions higher than 2V03, each axis is numbered for each module.

2. *The closer the base board is to CPU, the sooner it will be numbered in the case there are several base boards.*

   In a general PC or IPC system, PCI slot is numbered from the one that is closest to CPU. However, there can be a system that applies number in the opposite way or in a random way.

3. *The numbering order in the same base board is J1,J2,J3, and J4.*

Allotting axis number, channel number

PCI slot #1 : BPFR(Board Num : 0)	
SIO-DB32(SUB1) : J1 connector ModuleNum: 0 ContactNum(DI/DO): 0~15/0~15	**SIO-DI32(SUB3) : J3 connector** ModuleNum: 2 ContactNum(DI): 16 ~ 47
SIO-DO32(SUB2) : J2 connector ModuleNum: 1 ContactNum(DO): 16~47	**SIO-DI32(SUB4) : J4 connector** ModuleNum: 3 ContactNum(DI): 48 ~ 79

PCI slot #2 : PCI-N804(Board Num : 1)
SMC-4V04(SUB1) : Embeded(J1/J2 connector) ModuleNum: 0 AxisNum : 0~3
SMC-4V04(SUB2) : Embeded(J3/J4 connector) ModuleNum: 1 AxisNum : 4~7

PCI slot #3 : PCI-R1604(Board Num : 3)	
RTEX-DI32(NODE0) ModuleNum: 7 ContactNum(DI): 112~143	**RTEX-PM(NODE16)** ModuleNum: 3 AxisNum: 9
RTEX-DO32(NODE1) ModuleNum: 8 ContactNum(DO): 128~159	**SIO-DI32(NODE17)** ModuleNum: 9 ContactNum(DI): 144 ~ 175
RTEX-PM (NODE2) ModuleNum: 2 AxisNum : 8	**A4N(NODE18)** ModuleNum: 4 AxisNum: 10
. . . Node number remaining empty.	. . . Node number remaining empty.

PCI slot #4 : BPHD(Board Num : 2)	
SIO-DB32(SUB1) ModuleNum: 4 ContactNum(DI): 80~95/48~79	**SIO-DB32(SUB3) : Embeded(J3 connector)** ModuleNum: 6 ContactNum(DI/DO): 96~111/112~127
SIO-DO32(SUB2) ModuleNum: 5 ContactNum(DO): 80~111	**SIO-DI32(NODE17)** ModuleNum: 9 ContactNum(DI): 144 ~ 175

PCI slot #5 : PCI-R1604-MLII(Board Num : 4)
SGDV (NODE ADDR : 0x41) On Net#1 ModuleNum: 5 AxisNum: 11
SGDV (NODE ADDR : 0x41) On Net#2 ModuleNum: 6 AxisNum: 12
JEPMC-IO2330 (NODE ADDR : 0x61) on NET#1 ModuleNum: 10 ContactNum(DI/DO): 176~239/160~223
JEPMC-PL2910 (NODE ADDR : 0x61) on NET#2 ModuleNum: 7 AxisNum: 13~14
. . . Node number remaining empty.

Base Command Manual Information

The manual is needed to run BIHR/BIFR/BPHR/BPFR/BC3R/BC6R/BPHD Base Board in Windows 98, Windows NT, Windows 2000 or Windows XP OS by Microsoft VC++6.0, Visual Basic, Borland C-Builder, and Delphi language. Library APIs included are described by each function.

Header File

C++

AXL.h

Visual Basic

AXL.bas

Delphi

AXL.pas

Terms for API

API Names in the Manual

API name used in this manual can be distinguished by Prefix for API's operation.

※ *Prefix for library API*

Axl : It means that API is only for base board and library. APIs starting with Axl are defined in AXL.h.

Set : It means that API sets register or variable of chip.

Get : It means that API checks variable value set by Set API in pair with Set, or read chip register state.

Argument Names in the Manual

The common arguments for APIs used in this manual have the following meanings.

long lBoardNo : The arrangement of board in ascending order from the first board of the initialized Base Board takes place. Board number starts from '0'.

DWORD uLevel : uLevel sets log level to be transmitted to EzSpy.

Base Command Quick List

Library Initialization

Function	Description
AxlOpen	Initialize library
AxlClose	Exit library use
AxlIsOpened	Check library initialization

Information on Library and Base Board

Function	Description
AxlGetBoardCount	Check the number of registered base board
AxlGetLibVersion	Check library version
AxlGetModuleNodeStatus	Check the connection between the slave nodes in the network

Base Command Function List

Library Initialization

AxlOpen

AxlOpenNoReset

AxlClose

AxlIsOpened

AxlInterruptEnable

AxlInterruptDisable

Information on Library and Base Board

AxlGetBoardCount

AxlGetLibVersion

AxlGetModuleNodeStatus

Library Initialization

Function	Description
AxlOpen	Initialize library.
AxlOnpenNoReset	Exit library use.
AxlClose	Exit library use.
AxlIsOpened	Check library initialization.
AxlInterruptEnable	Board Interrupt Enable.
AxlInterruptDisable	Board Interrupt Disable.

AxlOpen

Purpose

Initialize library.

Format

C++

```
DWORD AxlOpen(long lIrqNo)
```

Visual Basic

```
Function AxlOpen(ByVal lIrqNo As Long) As Long
```

Delphi

```
function AxlOpen(lIrqNo : LongInt) : DWord; stdcall;
```

Parameters

[in/out] Name	[Init Value] Explanation
[>>]lIrqNo	IRQ No. (0 ~ 64)

Return Values

[0000] AXT_RT_SUCCESS : API function executed successfully
[1001] AXT_RT_OPEN_ERROR : Library is not open
[1002] AXT_RT_OPEN_ALREADY : Library is already open
* See error code Table for more information on status error codes

Description

When library is initialized and user uses interrupt, registering the number of IRQ that processes Interrupt makes it available to use interrupt. For PCI and cPCI bus that supports PnP function, as the IRQ number is automatically allotted, the IRQ number set by user is neglected.

For ISA bus, the IRQ number can be taken from the value of Irq setting Dip S/W in Base Board. For PCI or cPCI bus, as they support PnP function, any value can be taken as nIrqNo is automatically allotted. The initial value when product is shipped out is set as Irq 7.

To use a library that Ajinextek provides, AxlOpen API must be called up in the first place. After using the library, allotted memory must be released by call up AxlClose API. Or else, system failure could occur out of memory leak.

Example

C++

```
#include "AXL.h"       // Add integrated library.
#include "AXHS.h"

// Initialize library.
// 7 means IRQ. IRQ is automatically set by PCI.
if (AxlOpen(7) == AXT_RT_SUCCESS)
  AfxMessageBox("Library has been initialized.");
```

```
else
  AfxMessageBox("Library has not been initialized.");
```

Visual Basic

```
' Add AXHS.bas and AXL.bas to use integrated library.

' Initialize library.
' 7 means IRQ. IRQ is automatically set by PCI.
If AxlOpen(7) = AXT_RT_SUCCESS Then
  MsgBox "Library has been initialized.", vbOKCancel
Else
  MsgBox "Library has not been initialized.", vbOKCancel
End If
```

Delphi

```
Uses
AXL, AXHS     { Add integrated library. }

{ Initialize library. }
{ 7 means IRQ. IRQ is automatically set by PCI. }
if (AxlOpen(7) = AXT_RT_SUCCESS) then
  Application.MessageBox('Library has been initialized.', 'Ajinextek', MB_OK)
else
  Application.MessageBox('Library has not been initialized.', 'Ajinextek', MB_OK);
MessageBox
```

See Also

AxlOnpenNoReset, AxlClose, AxlIsOpened

AxlOpenNoReset

Purpose

Open library without the process of initialization.

Format

C++

```
DWORD AxlOpenNoReset(long lIrqNo);
```

Visual Basic

```
Function AxlOpenNoReset(ByVal lIrqNo As Long) As Long
```

Delphi

```
function AxlOpenNoReset (lIrqNo : LongInt) : DWord; stdcall;
```

Parameters

[in/out] Name	[Init Value] Explanation
[>>]lIrqNo	IRQ 번호

Return Values

[0000] AXT_RT_SUCCESS : API function executed successfully
[1001] AXT_RT_OPEN_ERROR : Library is not open
[1002] AXT_RT_OPEN_ALREADY : Library is already open
* See error code Table for more information on status error codes

Description

This function opens the library without the process of initialization. This function also registers the IRQ number so that users can use the interrupt related functions if users want. As IRQ number is automatically assigned in case of PCI bus or cPCI bus that support PnP function, the IRQ number given by the user will be ignored.

For ISA bus product, IRQ number is configured by the DIP switches on the base board, and the DIP switch settings for PCI bus and cPCI bus will be ignored since they support PnP function. The factory default setting of IRQ number is 7.

The AxlOpen function must called in order to use the library provided by our company. And the AxlClose function must be called after finishing using the library to release the allocated memory, otherwise memory leak might cause system failure.

Examples

C++

```
#include "AXL.h"       // Add integrated library.
#include "AXHS.h"

// Initialize library.
// 7 means IRQ. IRQ is automatically set by PCI.
if (AxlOpenNoReset(7) == AXT_RT_SUCCESS)
```

```
  AfxMessageBox("Library has been initialized.");
else
  AfxMessageBox("Library has not been initialized.");
```

Visual Basic

```
' Add AXHS.bas and AXL.bas to use integrated library.

' Initialize library.
' Parameter '7' is IRQ number. IRQ is automatically set in case of PCI product.
If AxlOpenNoReset(7) = AXT_RT_SUCCESS Then
  MsgBox "Library has been initialized.", vbOKCancel
Else
  MsgBox "Library has not been initialized.", vbOKCancel
End If
```

Delphi

```
uses
AXL, AXHS     { Add integrated library. }

{ Initialize library. }
{ Parameter '7' is IRQ number. IRQ is automatically set in case of PCI product. }
if (AxlOpenNoReset(7) = AXT_RT_SUCCESS) then
  Application.MessageBox('Library has been initialized.', 'Ajinextek', MB_OK)
else
  Application.MessageBox('Library has not been initialized.', 'Ajinextek', MB_OK);
```

See Also

AxlOpen, AxlClose, AxlIsOpened

AxlClose

Purpose

Exit library use.

Format

C++

```
BOOL AxlClose();
```

Visual Basic

```
Function AxlClose() As Byte
```

Delphi

```
function AxlClose( ): Boolean; stdcall;
```

Description

Exit library use. As the allotted memory is released, call it up when library use is exited. Initialize it to use it again.

Return Values

[0] AXL library fails to shutdown
[1] AXL library shutdown successfully

Example

C++

```
// Exit library use.
AxlClose();
```

Visual Basic

```
' Exit library use.
AxlClose
```

Delphi

```
{ Exit library use. }
AxlClose();
```

See Also

AxlOpen, AxlClose, AxlIsOpened

AxlIsOpened

Purpose

Check library initialization.

Format

C++

```
BOOL AxlIsOpened();
```

Visual Basic

```
Function AxlIsOpened() As Byte
```

Delphi

```
function AxlIsOpened():Boolean; stdcall;
```

Description

Check library initialization and whether it is usable. If API is used in the state that library has not been initialized, various problems can happen. Thus, check library initialization by using the API before using library.

Return Values

[0] AXL library does not initialize
[1] AXL library initialized

Example

C++

```
// Check library initialization.
if(AxlIsOpened())
  AfxMessageBox("Library has been initialized.");
else
  AfxMessageBox("Library has not been initialized.");
```

Visual Basic

```
' Check library initialization.
If AxlIsOpened Then
  MsgBox "Library has been initialized.", vbOKCancel
Else
  MsgBox "Library has not been initialized.", vbOKCancel
End If
```

Delphi

```
{ Check library initialization. }
if(AxlIsOpened()) then
  Application.MessageBox('Library has been initialized.', 'Ajinextek', MB_OK);
else
  Application.MessageBox('Library has not been initialized.', 'Ajinextek', MB_OK);
```

See Also

AxlOpen, AxlClose

AxlInterruptEnable

Purpose

To enable on-board interrupt.

Format

C++

```
DWORD AxlInterruptEnable();
```

Visual Basic

```
Function AxlInterruptEnable() As Long
```

Delphi

```
function AxlInterruptEnable() : DWord; stdcall;
```

Return Values

[0000] AXT_RT_SUCCESS : API function executed successfully

* See error code Table for more information on status error codes

Description

This function must be called first in order to use on-board interrupt.

Examples

C++

```
// USE WINDOW MESSAGE TO RECEIVE INTERRUPT MESSAGE
BEGIN_MESSAGE_MAP(CCAMCIPDlg, CDialog)
   //}}AFX_MSG_MAP
   // DECLARATION
   ON_MESSAGE(WM_CAMCIP_INTERRUPT, OnTriggerInterrupt)
END_MESSAGE_MAP()

AxlInterruptEnable();
// 'OnTriggerInterrupt()' FUNCTION WILL BE CALLED WHEN INTERRUPT OCCURS
AxmInterruptSetAxis(0, m_hWnd, NULL, NULL);
```

Visual Basic

```
' USE WINDOW MESSAGE TO RECEIVE INTERRUPT MESSAGE
' DECLARATION
Private Sub AxtMsg1_OnMessage1(ByVal wParam As Long, ByVal lParam As Long)
' INTERRUPT MESSAGE PROCESS SCRIPT
End Sub

' 'AxtMsg1_OnMessage1()' FUNCTION WILL BE CALLED WHEN INTERRUPT OCCURS
AxtMsg1.Message1 = WM_CAMCIP_INTERRUPT
AxlInterruptEnable
' 'OnTriggerInterrupt()' WILL BE CALLED WHEN INTERRUPT OCCURS
AxmInterruptSetAxis 0, Me.hWnd, WM_CAMCIP_INTERRUPT, 0
```

Delphi

```
{ USE WINDOW MESSAGE TO RECEIVE INTERRUPT MESSAGE. }
{ DECLARATION }
procedure TForm1.OnInterruptMessage(var Msg : TMessage);
begin
  { INTERRUPT MESSAGE PROCESS SCRIPT }
  { 'OnInterruptMessage()' FUNCTION WILL BE CALLED WHEN INTERRUPT OCCURS }
  AxlInterruptEnable();
  AxmInterruptSetAxis(Form1.Handle, WM_CAMCIP_INTERRUPT, nil);
end;
```

See Also

AxlInterruptDisable

AxlInterruptDisable

Purpose

To disable on-board interrupt.

Format

C++

```
DWORD AxlInterruptDisable();
```

Visual Basic

```
Function AxlInterruptDisable() As Long
```

Delphi

```
function AxlInterruptDisable() : DWord; stdcall;
```

Return Values

[0000] AXT_RT_SUCCESS : API function executed successfully

* See error code Table for more information on status error codes

Description

Call this function to disable the interrupt if the AxlInterruptEnable function was used to enable the interrupt.

Examples

C++

```
// Interrupt must be disabled after using the interrupt.
AxlInterruptDisable();
```

Visual Basic

```
' Interrupt must be disabled after using the interrupt.
AxlInterruptDisable
```

Delphi

```
{ Interrupt must be disabled after using the interrupt. }
AxlInterruptDisable();
```

See Also

AxlInterruptEnable

Information on Library and Base Board

Function	Description
AxlGetBoardCount	Check the number of registered base board.
AxlGetLibVersion	Check library version.
AxlGetModuleNodeStatus	Check the connection between the slave nodes in the network.

AxlGetBoardCount

Purpose

Check the number of registered base board.

Format

C++

```
DWORD AxlGetBoardCount(long *lpBoardCounts);
```

Visual Basic

```
Function AxlGetBoardCount(ByRef lpBoardCounts As Long) As Long
```

Delphi

```
function AxlGetBoardCount(lpBoardCounts : PLongInt) : DWord; stdcall;
```

Parameters

[in/out] Name	[Init Value] Explanation
[<<]lpBoardCounts	Number of base board

Return Values

[0000] AXT_RT_SUCCESS : API function executed successfully
[1053] AXT_RT_NOT_OPEN : Failed to initialize the library
* See error code Table for more information on status error codes

Description

Check the number of registered base board.

Examples

C++

```
// Check the number of registered base board.
long lBoardCounts;
CString strData;

AxlGetBoardCount(&lBoardCounts);
strData.Format("The number of entire registered base board is %d.", lBoardCounts);

AfxMessageBox(strData);
```

Visual Basic

```
' Check the number of registered base board.
Dim lBoardCounts As Long
Dim strData As String

AxlGetBoardCount(lBoardCounts)
strData = "The number of entire registered base board is" + CStr(lBoardCounts) + "."
MsgBox strData
```

Delphi

```
{ Check the number of registered base board. }
var
   lBoardCounts : LongInt;
   strData : String;

begin
   AxlGetBoardCount(@lBoardCounts);
   strData := 'The number of entire registered base board is ' + IntToStr(lBoardCounts)
      + '.';
   Application.MessageBox (PCHAR(strData), 'Ajinextek', MB_OK);
end;
```

See Also

AxlGetLibVersion

AxlGetLibVersion

Purpose

Check library version.

Format

C++

```
DWORD AxlGetLibVersion(char *szVersion);
```

Visual Basic

```
Function AxlGetLibVersion(ByRef szVersion As Byte) As Long
```

Delphi

```
function AxlGetLibVersion(szVersion : PChar) : DWord; stdcall;
```

Parameters

[in/out] Name	[Init Value] Explanation
[>>]szVersion	Library version

Return Values

[0000] AXT_RT_SUCCESS : API function executed successfully
[1053] AXT_RT_ERROR_VERSION_READ : Failed to initialize the library
* See error code Table for more information on status error codes

Description

Check library version. As library is updated regularly, you can check the version information to decide whether to update it or not.

Examples

C++

```
// Check library version.
char szVersion[50];
CString strData;

AxlGetLibVersion(szVersion);
strData.Format("The current library version is %s.", szVersion);

AfxMessageBox(strData);
```

Visual Basic

```
' Check library version.
Dim byVersion(30) As Byte

AxlGetLibVersion byVersion(0)
MsgBox StrConv(CStr(byVersion), vbUnicode)
```

Delphi

```
{ Check library version. }
var
   strData : String;
   szVersion : array [0..29] of Char;

begin
   AxlGetLibVersion(@szVersion);
   strData := ' The current library version is ' + PCHAR(@szVersion)+ '.';
   Application.MessageBox(PCHAR(strData), 'Ajinextek', MB_OK);
end;
```

See Also

AxlGetBoardCount

AxlGetModuleNodeStatus

Purpose

Check the connection between the slave nodes in the network.

Format

C++

```
DWORD AxlGetModuleNodeStatus(long lBoardNo, long lModulePos);
```

Visual Basic

```
Function AxlGetModuleNodeStatus(ByVal lBoardNo As Long, ByVal
lModulePos As Long) As Long
```

Delphi

```
function AxlGetModuleNodeStatus(lBoardNo : LongInt; lModdulePos :
LongInt) : DWord; stdcall;
```

Parameters

[in/out] Name	[Init Value] Explanation
[>>]lBoardNo	Setting the base board number. Number is assigned in order of slot from 0.
[>>]lModulePos	Setting the relative position of the slave node number which associated with the master board. - PCI-R1604 : Node address start from 0 to 31 - PCI-R1604-MLII : in NET#1the node address start from 0 to 29

Return Values

[0000] AXT_RT_SUCCESS : API function executed successfully
[1054] AXT_RT_NOT_SUPPORT_VERSION : Hardware not supported
[1101] AXT_RT_INVALID_BOARD_NO : Invalid board number
[1102] AXT_RT_INVALID_MODULE_POS : Invalid module position
[1152] AXT_RT_NETWORK_ERROR : Slave nodes is not connected to the network or connection is faulty
[1153] AXT_RT_NETWORK_LOCK_MISMATCH : Stored connection information and the actual connection information is incorrect. Lock of the board in EzConfig information does not match with the actual connection status. Lock information for reconstruction of a slave existing Ezconfig should give clear in the Unlock

* See error code Table for more information on status error codes

Description

Check the connection status that the slave node connected to the master board. Return the value of the function AXT_RT_SUCCESS(0000) means that the connection is normally.

Examples

C++

```
// Check the connection status.
DWORD dwReturn;
dwReturn = AxlGetModuleNodeStatus (0, 0);
if(dwReturn)
   strData.Format("The connection of 0th node has problem : %d.", dwReturn);
else
   strData.Format("The connection of 0th node is normally.");
AfxMessageBox(strData);
```

Visual Basic

```
' Check the connection status.
If AxlGetModuleNodeStatus (0,0) = AXT_RT_SUCCESS Then
   MsgBox "The connection of 0th node has problem.", vbOKCancel
Else
   MsgBox "The connection of 0th node is normally.", vbOKCancel
End If
```

Delphi

```
{ Check the connection status. }
var
   dwReturn : DWord;
   lBoardCounts : LongInt;
   strData : String;

begin
   dwReturn = AxlGetModuleNodeStatus (0,0);
   if (dwReturn )then
      strData := 'The connection of 0th node has problem.';
   else
      strData := 'The connection of 0th node is normally.';
end;
```

See Also

AxlGetBoardCount

Error Code Table Check

Error coded returned from the library functions can be used to check the errors generated during the execution of library functions and to process the result accordingly. Error code returned from each of the library functions indicates reasons of failure of corresponding function, and you can check the error code to understand the cause of failure. In general, it is good idea to check the code when the function returns the code other than AXT_RT_SUCCESS for the purpose of easier debugging.

Error Code	Description
[0000] AXT_RT_SUCCESS	API function executed successfully
[1001] AXT_RT_OPEN_ERROR	Library is not open
[1002] AXT_RT_OPEN_ALREADY	Library is already open
[1053] AXT_RT_NOT_OPEN	Failed to initialize the library
[1054] AXT_RT_NOT_SUPPORT_VERSION	Hardware not supported
[1101] AXT_RT_INVALID_BOARD_NO	Invalid board number
[1102] AXT_RT_INVALID_MODULE_POS	Invalid module position
[1103] AXT_RT_INVALID_LEVEL	Invalid level
[1151] AXT_RT_ERROR_VERSION_READ	Failed to read library version
[1152] AXT_RT_NETWORK_ERROR	Slave nodes is not connected to the network or connection is faulty
[1153] AXT_RT_NETWORK_LOCK_MISMATCH	Stored connection information and the actual connection information is incorrect. Lock of the board in EzConfig information does not match with the actual connection status. Lock information for reconstruction of a slave existing Ezconfig should give clear in the Unlock
[1160] AXT_RT_1ST_BELOW_MIN_VALUE	First parameter is below the minimum value
[1161] AXT_RT_1ST_ABOVE_MAX_VALUE	First parameter is above the maximum value
[1170] AXT_RT_2ND_BELOW_MIN_VALUE	Second parameter is below the minimum value
[1171] AXT_RT_2ND_ABOVE_MAX_VALUE	Second parameter is above the maximum value
[1180] AXT_RT_3RD_BELOW_MIN_VALUE	Third parameter is below the minimum value
[1181] AXT_RT_3RD_ABOVE_MAX_VALUE	Third parameter is above the maximum value
[1190] AXT_RT_4TH_BELOW_MIN_VALUE	Fourth parameter is below the minimum value
[1191] AXT_RT_4TH_ABOVE_MAX_VALUE	Fourth parameter is above the maximum value
[1200] AXT_RT_5TH_BELOW_MIN_VALUE	Fifth parameter is below the minimum value
[1201] AXT_RT_5TH_ABOVE_MAX_VALUE	Fifth parameter is above the maximum value
[1210] AXT_RT_6TH_BELOW_MIN_VALUE	Sixth parameter is below the minimum value
[1211] AXT_RT_6TH_ABOVE_MAX_VALUE	Sixth parameter is above the maximum value
[1220] AXT_RT_7TH_BELOW_MIN_VALUE	Seventh parameter is below the minimum value
[1221] AXT_RT_7TH_ABOVE_MAX_VALUE	Seventh parameter is above the maximum value

[1230] AXT_RT_8TH_BELOW_MIN_VALUE	Eighth parameter is below the minimum value
[1231] AXT_RT_8TH_ABOVE_MAX_VALUE	Eighth parameter is above the maximum value
[1240] AXT_RT_9TH_BELOW_MIN_VALUE	Nineth parameter is below the minimum value
[1241] AXT_RT_9TH_ABOVE_MAX_VALUE	Nineth parameter is above the maximum value
[1250] AXT_RT_10TH_BELOW_MIN_VALUE	Tenth parameter is below the minimum value
[1251] AXT_RT_10TH_ABOVE_MAX_VALUE	Tenth parameter is above the maximum value
[2001] AXT_RT_AIO_OPEN_ERROR	Failed to open AIO module
[2051] AXT_RT_AIO_NOT_MODULE	No AIO module
[2052] AXT_RT_AIO_NOT_EVENT	Failed to read AIO event
[2101] AXT_RT_AIO_INVALID_MODULE_NO	Invalid AIO module
[2102] AXT_RT_AIO_INVALID_CHANNEL_NO	Invalid AIO channel number
[2106] AXT_RT_AIO_INVALID_USE	Can not use AIO
[2107] AXT_RT_AIO_INVALID_TRIGGER_MODE	Invalid trigger mode
[3001] AXT_RT_DIO_OPEN_ERROR	Failed to open DIO module
[3051] AXT_RT_DIO_NOT_MODULE	No DIO module
[3052] AXT_RT_DIO_NOT_INTERRUPT	DIO Interrupt not set
[3101] AXT_RT_DIO_INVALID_MODULE_NO	Invalid DIO module number
[3102] AXT_RT_DIO_INVALID_OFFSET_NO	Invalid DIO OFFSET number
[3103] AXT_RT_DIO_INVALID_LEVEL	Invalid DIO level
[3104] AXT_RT_DIO_INVALID_MODE	Invalid DIO mode
[3105] AXT_RT_DIO_INVALID_VALUE	Invalid setting value
[3106] AXT_RT_DIO_INVALID_USE	Can not use DIO API function
[4001] AXT_RT_MOTION_OPEN_ERROR	Failed to open motion library
[4051] AXT_RT_MOTION_NOT_MODULE	No motion module is installed in the system
[4052] AXT_RT_MOTION_NOT_INTERRUPT	Failed to read the resul of interrupt
[4053] AXT_RT_MOTION_NOT_INITIAL_AXIS_NO	Failed to initialize the motion of corresponding axis
[4054] AXT_RT_MOTION_NOT_IN_CONT_INTERPOL	Command to stop continuous interpolation while not in continuous interpolation motion
[4055] AXT_RT_MOTION_NOT_PARA_READ	Failed to load parameters for home return drive
[4101] AXT_RT_MOTION_INVALID_AXIS_NO	Corresponding axis does not exist
[4102] AXT_RT_MOTION_INVALID_METHOD	Invalid setting for corresponding axis drive
[4103] AXT_RT_MOTION_INVALID_USE	Invalid setting for 'uUse' parameter
[4104] AXT_RT_MOTION_INVALID_LEVEL	Invalid setting for 'uLevel' parameter
[4105] AXT_RT_MOTION_INVALID_BIT_NO	Invalid setting for general purpose input/output bit
[4106] AXT_RT_MOTION_INVALID_STOP_MODE	Invalid setting values for motion stop mode
[4107] AXT_RT_MOTION_INVALID_TRIGGER_MODE	Invalid setting for trigger setting mode

[4108] AXT_RT_MOTION_INVALID_TRIGGER_LEVEL	Invalid setting for trigger output level
[4109] AXT_RT_MOTION_INVALID_SELECTION	'uSelection' parameter is set to a value other than COMMAND or ACTUAL
[4110] AXT_RT_MOTION_INVALID_TIME	Invalid setting for Trigger output time value
[4111] AXT_RT_MOTION_INVALID_FILE_LOAD	Failed to load the file containing motion setting values
[4112] AXT_RT_MOTION_INVALID_FILE_SAVE	Failed to save the file containing motion setting values
[4113] AXT_RT_MOTION_INVALID_VELOCITY	Motion error occurred since the motion drive velocity value is set as zero
[4114] AXT_RT_MOTION_INVALID_ACCELTIME	Motion error occurred since motion drive acceleration time value is set as zero
[4115] AXT_RT_MOTION_INVALID_PULSE_VALUE	Input pulse value is set to a value less than zero when setting motion unit
[4116] AXT_RT_MOTION_INVALID_NODE_NUMBER	Called position or velocity override function while not in motion
[4117] AXT_RT_MOTION_INVALID_TARGET	Flag indicating the cause of multi axis motion stop
[4151] AXT_RT_MOTION_ERROR_IN_NONMOTION	Not in motion drive when it should be
[4152] AXT_RT_MOTION_ERROR_IN_MOTION	Called another motion drive fuction before the completetion of current motion
[4153] AXT_RT_MOTION_ERROR	Error occurred during the execution of multi axis motion stop function
[4154] AXT_RT_MOTION_ERROR_GANTRY_ENABLE	Gantry-enable is activated again while in motion with gantry enabled
[4155] AXT_RT_MOTION_ERROR_GANTRY_AXIS	Invalid input of gantry axis master channel (axis) number (starting from zero)
[4156] AXT_RT_MOTION_ERROR_MASTER_SERVOON	Servo ON is not enabled for the master axis
[4157] AXT_RT_MOTION_ERROR_SLAVE_SERVOON	Servo ON is not enabled for the slavve axis
[4158] AXT_RT_MOTION_INVALID_POSITION	Not in a valid position
[4159] AXT_RT_ERROR_NOT_SAME_MODULE	Not in the same module
[4160] AXT_RT_ERROR_NOT_SAME_BOARD	Not in the same board
[4161] AXT_RT_ERROR_NOT_SAME_PRODUCT	When products are different each other
[4162] AXT_RT_NOT_CAPTURED	Failed to capture the position
[4163] AXT_RT_ERROR_NOT_SAME_IC	Not in the same chip
[4164] AXT_RT_ERROR_NOT_GEARMODE	Failed to change to the gear mode
[4165] AXT_ERROR_CONTI_INVALID_AXIS_NO	Invalid axis during continuous interpolation axix mapping
[4166] AXT_ERROR_CONTI_INVALID_MAP_NO	Invalid mapping number during continuous interpolation mapping
[4167] AXT_ERROR_CONTI_EMPTY_MAP_NO	Continuous interpolation mapping number is empty
[4168] AXT_RT_MOTION_ERROR_CACULATION	Error in calculation
[4169] AXT_RT_ERROR_MOVE_SENSOR_CHECK	Continuous interpolation error sensor (Alarm, EMG, Limit, etc) is detected

[4170] AXT_ERROR_HELICAL_INVALID_AXIS_NO	Invalid axis for helical axis mapping
[4171] AXT_ERROR_HELICAL_INVALID_MAP_NO	Invalid mapping number for helical mapping
[4172] AXT_ERROR_HELICAL_EMPTY_MAP_NO	Helical mapping number is empty
[4180] AXT_ERROR_SPLINE_INVALID_AXIS_NO	Invalid axis for spline axis mapping
[4181] AXT_ERROR_SPLINE_INVALID_MAP_NO	Invalid mapping number for spline mapping
[4182] AXT_ERROR_SPLINE_EMPTY_MAP_NO	Spline Mapping number is empty
[4183] AXT_ERROR_SPLINE_NUM_ERROR	Spline point value is inapplicable
[4184] AXT_RT_MOTION_INTERPOL_VALUE	Invalid input value for interpolation
[4185] AXT_RT_ERROR_NOT_CONTIBEGIN	CONTIBEGIN function is not called in continuous interpolation
[4186] AXT_RT_ERROR_NOT_CONTIEND	CONTIEND function is not called in continuous interpolation
[4201] AXT_RT_MOTION_HOME_SEARCHING	Other function is called while in home search motion
[4202] AXT_RT_MOTION_HOME_ERROR_SEARCHING	Home search motion is fored to stop by the user or something from external
[4203] AXT_RT_MOTION_HOME_ERROR_START	Failed to start home search drive due to intialization problem
[4204] AXT_RT_MOTION_HOME_ERROR_GANTRY	Failed to execute Gantry enable during home search motion
[4251] AXT_RT_MOTION_POS_OUTOFBOUND	Configured position is either above the maximum value or below the minimum value
[4252] AXT_RT_MOTION_PROFILE_INVALID	Invalid setting for velocity profile
[4253] AXT_RT_MOTION_VELOCITY_OUTOFBOUND	Configured velocity is either above the maximum value or below the minimum value
[4254] AXT_RT_MOTION_MOVE_UNIT_IS_ZERO	Unit of the motion values is set as zero
[4255] AXT_RT_MOTION_SETTING_ERROR	Invalid setting for Velocity, Acceleration, Jerk, or Profile
[4256] AXT_RT_MOTION_IN_CONT_INTERPOL	Execuation of motion start or restart function before the completion of current continous interpolated motion
[4257] AXT_RT_MOTION_DISABLE_TRIGGER	Trigger output is disabled
[4258] AXT_RT_MOTION_INVALID_CONT_INDEX	Invalid setting for continuous interpolation index value
[4259] AXT_RT_MOTION_CONT_QUEUE_FULL	Continuous Interpolation queue of motion chip is full
[4260] AXT_RT_PROTECTED_DURING_SERVOON	Can not be used the function or input value in servo-on status
[4261] AXT_RT_HW_ACCESS_ERROR	H/w of board access error has occurred. Check the status of the board mounting system or battery operation to check the status of power input